philosophies

Article

Research on the Mathematical Principles of Chinese Philosophy from the Body Dimension in Traditional Chinese Medicine

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Abstract

Many scholars believe that the *Yi Jing* 易經 (the Book of Changes) and traditional Chinese medicine share common mathematical principles, which are both predicated on the ontological of *qi* 氣 and the cosmological of correlative between nature and human. Traditional Chinese medicine emphasizes the systemic organization of organs, meridians, *qi*, and blood as central components by incorporating the mathematical principles, including the theory of "Chaos-Crack", the infinite classification methods of *yinyang* 陰陽, the generative and restrictive interactions of *wuxing* 五行, and the metaphysical significance of special numbers such as one, two, three, etc. Traditional Chinese medicine also formulates many theories and methodologies by integrating these mathematical principles with the schemata of *luoshu* 洛書 and *jiugong* 九宮, as well as the special combination numbers such as *tianliu diwu* 天六地五. This research tries to explain the mathematical principles and applications from the body dimension in traditional Chinese medicine.

Keywords: *yinyang*; *wuxing*; mathematical principles; body dimension; Chinese philosophy; traditional Chinese medicine

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Academic Editor: Jenny Hung

Received: 8 August 2025
Revised: 3 October 2025
Accepted: 6 October 2025
Published: 8 October 2025

Citation: Xie, H.; Yan, R. Research on the Mathematical Principles of Chinese Philosophy from the Body Dimension in Traditional Chinese Medicine. *Philosophies* **2025**, *10*, 111. https://doi.org/10.3390/philosophies10050111

1. Introduction

Many scholars believe that traditional Chinese medicine and *Yi Jing* 易經 (the Book of Changes) share common mathematical principles. The classical texts of traditional Chinese medicine are often attributed to Huang Di 黄帝, who is regarded as a seminal figure in ancient Chinese mathematics alongside the *Yi Jing*. This association reflects the intrinsic connection between mathematics and traditional Chinese medicine. There are many discussions about the mathematical principles in the *Huang Di Nei Jing* 黄帝内經, particularly concerning the concepts of *zhishu* 至數 (the ultimate number) and *zhidao* 至道 (the ultimate rule). Qi Bo 歧伯, a legendary doctor in the history of traditional Chinese medicine, explained the *zhishu* as follows: "The perfect numbers of heaven and earth, they begin with one and they end with nine" 天地之至數, 始於一, 終於九焉 [1] (Vol. 1, p. 352). The numbers can infiltrate into the human body, significantly influencing the circulation of blood and *qi* 氣 (vital energy), and even affecting overall health.

The mathematical principles underlying traditional Chinese medicine represent a crucial connection that integrates heaven, earth, human, *yinyang* 陰陽 (two basic contrary forces), and *wuxing* 五行 (the five elements) into a coherent system. However, these principles remain both obscure and fragmented, resulting in insufficient scholarly attention. Although there are some relevant discussions [2–4], the analyses frequently prioritize medical over a thorough exploration of mathematical philosophy. As Professor Wen Xing analyzed, the core idea of Chinese mathematical philosophy is to use the mathematical principles and methods to investigate the relationship between nature and human society [5]. The consistency of this perspective throughout the evolution of ancient Chinese philosophy underscores its significance in addressing the opportunities of Chinese philosophy. By studying the mathematical principles in traditional Chinese medicine, novel insights can be derived, which can promote the development of traditional Chinese medicine philosophy. Since traditional Chinese medicine frequently uses numbers to elucidate the structure and function of the human body, this article chooses to explore the mathematical principles of the human body in traditional Chinese medicine from the perspective of Chinese mathematical philosophy.

The term "Chinese mathematical philosophy" refers to the theory proposed and analyzed by Professor Xing Wen, who is the Presidential Chair Professor of traditional Chinese art and culture at CUHK Shenzhen and the Robert 1932 and Barbara Black Professor in Asian Studies, Emeritus, Dartmouth College. Chinese mathematical philosophy is based on the mathematical thoughts within ancient Chinese cosmology, characterized by discontinuity and irregularity. It aims to explore the philosophical issue of the relationship between heaven and human through mathematical principles and methods. Professor Xing Wen categorizes the concept of *shu* 數 (numbers) in the ancient Chinese cosmology into three levels based on the differences in mathematical principles. The first level is *hundun zhishu* 混沌之數 (chaos number); the second level includes the *liangyi zhishu* 兩儀之數 (two forms numbers), and *sansheng zhishu* 三生之數 (three generative numbers); the third level comprises the numbers and principles of *sixiang* 四象 (four emblems), *bagua* 八卦 (eight trigrams), and *wanwu* 万物 (all things) [5].

Several important analytical methods and theories used in this article, such as the mathematical principle of *hundun-benglie* 混沌-崩裂 (Chaos-Crack) and the metaphysical numbers of "Chaos-One," "Crack-One," and "Entirety-One," are all based on the framework of Chinese mathematical philosophy. In addition, the main purpose of this research is to reveal and discuss the special *shu* in Chinese philosophy and the thoughts underlying them regarding the metaphysical *Dao* 道 (ultimate rule) of the universe, thereby demonstrating that *shu* in Chinese philosophy are not merely "numbers", but also a method and schema with metaphysical significance for understanding the *Dao* and grasping the relationship between nature and human. Therefore, this article did not discuss the connotations of natural numbers such as one, two, three, and four in sequence, but focused on their *shuli* 數理 (mathematical principles) and the relationships among them.

Additionally, the reason why this research chooses to start the discussion of *shuli* from the body dimension is that traditional Chinese medicine not only uses numbers to measure, analyze, and symbolize the human body's organs, data, and structures, but also views the mathematical principles as the crucial role for understanding laws of life, developing medical theories, and executing medical practices. The relevant understanding and knowledge of mathematical principles in traditional Chinese medicine forms the foundation for the development of medical theories and treatment methods.

2. Materials and Methods

The main concepts related to *shuli* are primarily recorded in the classic texts of Chinese philosophy, especially the *Yi Jing*, *Li Ji* 禮記, *Zhuang Zi* 莊子, *Huai Nan Zi* 淮南子, etc. Some of them are well known to Western scholars, such as the *shengshu* 生數 (the generative numbers) and the *chengshu* 成數 (the completed numbers) mentioned in the *Yi Jing*. Therefore, this article did not discuss them much. Meanwhile others, such as the *zhishu*, the numbers of meridians, and the mathematical principles of *yinyang* and *wuxing*, which are the key issues in the *Huang Di Nei Jing*, have attracted relatively little scholarly attention. Therefore, this article paid more attention to them.

The main materials used in this article are the classic texts of traditional Chinese medicine, especially the *Huang Di Nei Jing*. It consists of two parts: *Su Wen* 素問 and *Ling Shu* 靈樞. Paul U. Unschuld, a Professor and director of the Horst-Goertz Endowment Institute for the Theory, History, and Ethics of Chinese Life Sciences at Chartié-Medical University Berlin, translated them, respectively. This article tried not to impose an interpretation, especially those coming from Western terminologies, but rather lets the texts themselves unfold the meanings of *shuli*, frequently through quotations. Therefore, all quotations from *Huang Di Nei Jing* retained the original Chinese words and Prof. Paul U. Unschuld's translations.

In order to illustrate the metaphysical connotation of *shu* in Chinese philosophy as fully as possible, this research employed two basic approaches for an understanding of *shuli* in the Chinese Philosophy: the descriptive and historical, the paradigmatic and conceptual. The descriptive approach begins with the historical record of *shuli* in early texts, attempting to establish the notion of *shuli* which is a dynamic and changing system of thought and way of dealing with the nature and human. Although emphasizing historical records, the purpose of this research is not primarily historical. Instead, it is meant to articulate conceptual positions. This is the second main line of approach. In this sense, although grounded in Sinology, this research is intended to be philosophical.

3. Two Different Significances of Zhishu

From the standpoint of Chinese mathematical philosophy, the *zhishu* system in the *Huang Di Nei Jing* belongs to the nine-digit numerical system, which stands in contrast to the decimal system predominant during the Han Dynasty. Its essence lies in the construction of numbers from one to nine as cardinal numbers, thereby imbuing them with unique mathematical characteristics and two completely different significances, just parallel to the bifurcation of ancient Chinese mathematics into the realms of arithmetic and numerology [6]. One aspect is *jisuan zhishu* 計算之術 (the computational technique), which is used to solve specific computational problems, and characterized by complexity, constructiveness, and practicality. The other aspect pertains to *shuli zhidao* 數理之道 (the mathematical principle), which endeavors to explore foundational mathematical principles and methodologies, and be characterized by abstraction, logic, and mystique.

Qin Jiushao (秦九韶 1208–1261), a prominent mathematician in the Southern Song Dynasty (南宋 1127–1279), articulated a similar differentiation concerning the roles of numbers and classified them into great and small categories. The *shuli zhidao* denotes numbers of substantial importance that can establish a connection with the divine and resonate with human nature and destiny. Conversely, the *jisuan zhishu* refers to numbers of lesser significance that are utilized to manage practical affairs and analyze phenomena within the universe [7] (p. 1). This distinction aligns with Bertrand Russell's analysis, which posited that mathematics can be pursued in two divergent directions: one that leads "towards gradually increasing complexity" and the other towards "greater and greater abstractness and logical simplicity" [8] (p. 1)

The *jisuan zhishu* in the body dimension serves various functions to traditional Chinese medicine, including measurement, analysis, and symbolism. It is essential for assessing the human body, identifying acupuncture points, and calculating drug dosage ratios. For example, the *Huang Di Nei Jing Ling Shu* 黃帝內經靈樞 uses numerical values to establish the connection between the physical structure of bones and the abstract notion of meridians. It delineates that the passage of the *qi* covers 6 *cun* 寸during one breathing, and the passage of the *qi* covers 810 *zhang* 丈 [9] (p. 239).

Furthermore, the *shuli zhidao* plays a crucial role for understanding laws of life, developing medical theories, and executing medical practices. Many theories and methods of traditional Chinese medicine are intricately connected to it, such as the theory of *wuyun liuqi* 五運六氣 (five evolutionary phases and six climatic factors), which is a comprehensive theory used to explain the relationship between diseases and climate change. It mainly involves the theory of mutual generation and mutual restraint among *yinyang* and *wuxing*, as well as the changing patterns of six typical climates: *feng* 風 (wind), *re* 熱 (heat), *huo* 火 (fire), *shi* 溼 (dampness), *zao* 燥 (dryness), and *han* 寒 (cold) [10]. It also can be traced back to the basic numerical concepts of *yuanqi* 元氣 (the meta vital energy), the metaphysical numbers two and five, and the trinity of heaven, earth, and human.

While these two categories of numbers have occasionally developed independently, there have also been instances of convergence, which have collectively facilitated the advancement of the mathematical philosophy of traditional Chinese medicine. Nevertheless, this convergence has also presented some challenges, including obscuring the *shuli zhidao* with an aura of mystery and deviating it from its abstract simplicity.

4. Mathematical Principles of Numbers One and Three

The metaphysical number of one is the beginning of the *zhishu*, which contains the mathematical principle of *hundun benglie*, and serves as the foundation of the mathematical principles of the human body. As the *Huang Di Nei Jing Su Wen* 黃帝內經素問 states the following: "The extension of the great void is boundless; it is the basis of [all] founding and it is the principal [source] of [all] transformation. The myriad beings depend on the great void to come into existence." 太虛廖廓, 肇基化元, 萬物資始 [1] (Vol. 2, p. 177). It distills the origin of creation and the generative processes to the following sequence: *taixu-yuanqi-wanwu* (太虛-元氣-萬物). These three concepts share a common substance of *qi*, and their numerical representation are one, but the mathematical principles contained within them are distinctly different, as follows.

The *taixu* 太虛 (the great void) is the "Chaos-One" that represents the indivisible cosmic origin. It is the ultimate category which describes the state of "Chaos" that existed before the differentiation of the universe. It just parallels the concept of "Chaos" in Western philosophy and *taiji* 太極 (the great ultimate) in the *Yi Jing*, all of which express an inseparable one. *Qin Jiushao* explained the relationship among *taixu*, number one and infinity, as he stated: "The role of mathematics stems from *taixu* which can derive to the number one and has the characteristic of cycle infinitely." 其用本太虛生一, 而周流無窮 [7] (p. 1).

The *yuanqi* 元氣 is the "Crack-One" that represents the original matrix from which all things in the universe are formed. Even if the "Chaos-One" cracks apart, it can still be regarded as a form of one, specifically as a mathematical set and the "Crack-One." It possesses characteristics of divisibility and combinability, which allows it to be gathered, dispersed, counted or manipulated. The notion of *zhao* 肇 refers to the initiation of time, while the notion of *ji* 基 signifies the beginning of space. Thereby, *yuanqi*, as evolved by *zhao* and *ji*, represents both the *qi* of material essence and one of mathematical philosophy, and contains the interpretations of *qi*, number, time, space, substance, and function.

The myriad beings are the "Entirety-One" that represents the unity of all existence and the holistic relationship between heaven and earth. The *qi* serves as the driving force that sustains the movement and transformation of all creatures. All things in the universe emerge from the transformation of *qi*, develop through its circulation, and ultimately return to it. Human beings as the integral components of the universe are also intricately connected to *qi*. Consequently, an individual's physical and mental well-being, and life trajectory, are significantly influenced by the prosperity or decline of *qi*. The processes of birth, aging, illness, and death are intricately linked to the state of *qi*.

The cosmological theories in the pre-Qin period led to two distinct types of mathematical interpretations, one type centered around the *liangyi zhishu* in the *Yi Jing* that emphasizes evolutionary roles of the number two and the mathematical principles of *yinyang* bifurcation; another type centered around *sansheng zhishu* in the *Dao De Jing* 道德經that emphasizes the evolutionary roles of the number three and the mathematical principles of triadic division and generation. Liu Xin (劉歆 50 BCE–23 CE), a famous mathematician during the Han Dynasty (漢朝 202 BC–220 AD), attributed the *liangyi zhishu* to the metaphysical number two, and the *sansheng zhishu* to the metaphysical number one [11] (p. 896).

In the traditional Chinese medicine philosophy, the number three and the mathematical principles underlying it are influenced both by the theory of *sancai* 三才 (the three elements of heaven, earth, and human) in the *Yi Jing*, as well as *sanfen sunyifa* 三分損益法 (the trisection profit and loss method) widely utilized during the Han Dynasty. The mathematical characteristic of unity among the *sancai* supports the integrated, systemic, and interactive holistic perspective of traditional Chinese medicine.

Additionally, many concepts in traditional Chinese medicine are influenced by the *sanfen sunyifa* which emphasizes differentiation, such as the theory of *sanyin sanyang* 三陰三陽, which is a special theory of traditional Chinese medicine based on the theory of *yinyang* in Chinese philosophy. It shows that the movement and change in *yin qi* and *yang qi* can produce six new states, respectively, the *taiyang* 太阳 (major *yang*), *yangming* 阳明 (*yang* brilliance), *shaoyang* 少阳 (minor *yang*), *taiyin* 太阴 (major *yin*), *shaoyin* 少阴 (minor *yin*), and *jueyin* 厥阴 (ceasing *yin*) [1] (Vol. 1, pp. 130–131).

Therefore, the metaphysical number three in traditional Chinese medicine contains two main mathematical principles. The first mathematical principle is *he'er weiyi* 合而为一 (multitudinous integrated into one), which is exemplified by the cosmic perspective of the correspondence between heaven and human, as well as the comprehension of the unity of *sancai*. Traditional Chinese medicine draws upon ancient Chinese philosophy and offers medicalized, embodied, and mathematical interpretations of the interconnectedness of all phenomena. Traditional Chinese medicine elucidates the static relationships between numerical representations of nature and the human body through various analytical frameworks, including numerical correspondence, typological proximity, morphological resemblance, and visual alignment.

Additionally, it underscores the significance of numbers in quantification and its relevance in guiding medical practices. Traditional Chinese medicine also asserts that the laws governing nature and human are dynamically interconnected through the concept of *qi*. An individual can comprehensively understand the dynamic interplay of their *qi* by quantifying the numbers of nature and human. Thereby, *qi* and numbers become two distinct manifestations of the principles that govern the relationship between nature and human, and make them to be the schemas that can be measurable, repeatable, or visualizable. As Richard E. Palmer analyzed: "Numbers are the cleanest, clearest, and abstractest of ideas, so they are especially prized. Then comes everything that is measurable, repeatable, or visualizable as schema." [12] (p. 225)

Another mathematical principle is *yifen weisan* 一分为三 (one divides into three), which is exemplified by *sansheng zhishu* and *sanfen sunyifa*. The significance of the number three in the Han Dynasty philosophy is associated with the idea of *sansheng wanwu* 三生万物 (three derives all things in the universe) in the *Dao De Jing*, as well as the *sanfen sunyifa*, which are widely utilized in the fields of Chinese music, astronomy, and medicine. Many scholars in the Han Dynasty placed considerable emphasis on these mathematical principles. For example, Liu Xin stated a special idea that "*Taiji and yuanqi* shows that the number one is merged by the metaphysical number of three" 太極元氣, 函三为一 [11] (p. 897). It means that *taiji and yuanqi* is the fundamental essence of all phenomena, it contains a developmental process characterized by stages defined by units of three [13] (p. 119). Traditional Chinese medicine is also significantly influenced by this principle, such as that it categorizes the human body into three sections: the upper, middle, and lower regions; each section is further subdivided into three indicators of heaven, earth, and human. Thereby, traditional Chinese medicine proposes a diagnostic method based on three sections and nine indicators.

The concept of *sanyin sanyang* in traditional Chinese medicine elucidates the intrinsic relationship between the numbers of *sansheng* and *liangyi*. Although these two types of number have distinct origins and differing mathematical properties, their interrelation can be elucidated through the *dayan zhishu* 大衍之數 (the great expansion numbers). This intrinsic connection facilitated the amalgamation during the Han Dynasty and contributed to the formation of the concept of *sanyin sanyang*. Although *yinyang* and the number of *liangyi* typically adhere to a binary division, *Huang Di Nei Jing* divides them into three.

5. Mathematical Principles of Yinyang and Wuxing

The mathematical principles of *liangyi zhishu*, and the bisection method of *yinyang* were endowed with new characters of infinite divisions and symmetrical coupling when the ideologists of traditional Chinese medicine used them to measure and treat the human body. In this context, *yin* and *yang* represent both the invisible *qi* at the ontological dimension and the differentiation numbers at the epistemological dimension. The concepts of the *sishi* 四時 (four seasons), *liuqi* 六氣 (six pneuma), and *bafeng* 八風 (eight winds) in nature, as well as the organs, meridians, and acupuncture points in the human body, are all divided by *yin* and *yang*. It reminds us of the *yichi zhichui* 一尺之錘 (a one-foot hammer) in the book of *Zhuangzi*, which exemplifies infinite divisibility and strict hierarchical structures, as it stated the following: "A one-foot hammer long, if you take half of it each day, will never be exhausted for all eternity." 一尺之錘, 日取其半, 萬世不竭 [14] (p. 298).

If we categorize the human body to *yin* and *yang* at the first level, the interior is classified as *yin*, while the exterior is classified as *yang*. The second level introduces further subdivisions. For example, the *wuzang* 五臟 (five viscera) and *liufu* (six bowels 六腑) are inside the human body, which should be categorized as *yin* at the first level. However, the second level presents a new classification: *wuzang* are classified as *yin*, and the *liufu* are classified as *yang*. Just as *Huang Di Nei Jing Su Wen* stated: "Speaking of the *yin* and *yang* of man, then the outside is *yang*, the inside is *yin*." 言人之陰陽, 則外為陽, 內為陰. "Speaking of the *yin* and *yang* among the depots and palaces of the human body, then the depots are *yin* and the palaces are *yang*." 言人身之藏府中陰陽,則藏者為陰, 府者為陽 [1] (Vol. 1, p. 89). By utilizing this methodology, the human body can be further classified into an infinitely and multi-layered system of *yinyang* categorizations.

This bisection method of *yinyang* fundamentally belongs to the classification techniques of formal logic. Its distinguishing characteristic is not the repetitive application of binary numerical principles; rather, it is the dialectical synthesis of absolute determinacy and relative variability. The organs, functions, and structure of the human body can be classified as either *yin* or *yang*. This classification upholds a general absolute determinacy, provided that consistent reference conditions are maintained. However, if the reference conditions change, it can result in localized relative variability. If there be a further breakdown of time or alterations in spatial environments, it will cause new classifications of *yinyang* in the human body [15] (pp. 31–37). Consequently, traditional Chinese medicine underscores the method of *sanyin zhiyi* 三因制宜 (the three factors for treatment), which advocates for the adoption of different therapeutic approaches based on temporal, locational, and individual variances.

In contrast to the *yinyang*, the *wuxing* incorporates a broader range of elements, which facilitates more complex classifications, combinations, and transformations. The heightened complexity is more effectively aligned with the systematic and variable demands inherent in traditional Chinese medicine. Consequently, it has emerged as another important fundamental theory, and reinforces the significance and distinctiveness of the number five and the mathematical principles associated with it. Since the mathematical principles in the *Yi Jing* are integrated into the *zhishu*, the numbers one, two, three, and four are classified as *shengshu*, whereas six, seven, eight, and nine are classified as the *chengshu*. The number five, which serves as the central number, is characterized by the dual meanings of generation and completion. As the Song Dynasty scientist Shen Kuo (沈括 1031–1095) summarized: "Only the *Huang Di Nei Jing Su Wen* states that the number five belongs to both the *shengshu* and the *chengshu*." 唯黃帝素問土生數五, 成數亦五 [16] (p. 62).

From an epistemological perspective, traditional Chinese medicine posits the *wuxing* as the foundational framework for categorizing all things. This classification includes various components of the human body, such as the organs, pathologies, and visceral systems, all of which correspond to the mathematical principles associated with the *wuxing*. From an ontological standpoint, the *wuxing* are interpreted as five distinct forms of movement of *qi*, which permeate both the cosmos and the human body. As the *Huang Di Nei Jing Ling Shu* stated: "Between heaven and earth, and within the Six [dimensions of universal] unity, there is nothing that is outside of the number five, and man too corresponds to it. It is not such that there is only a *yin* and *yang* categorization." 天地之間, 六合之內, 不離於五, 人亦應之, 非徒一陰一陽而已也 [9] (p. 610).

Traditional Chinese medicine posits that the relationships of generation and restriction among the *wuxing* are characterized by nonlinearity and discontinuity. These relationships manifest as complementary cycles of restriction and generation, which encompass four derivative phenomena: I generate, generate me, I restrict, and restrict me. Traditional Chinese medicine also recognizes that the differences in material, quantity, and degree can result in distinct abnormalities in the generative and restrictive relationships among the *wuxing*. Thereby, traditional Chinese medicine constructed new theoretical constructs, including the *cheng* 乘 (overwhelming), *wu* 侮 (insulting), *sheng* 勝 (prevailing), and *fu* 復 (restoring), which enhanced the complexity, systemic, and dynamic equilibrium of the theory of *wuxing* [15] (pp. 44–49). The system of *wuxing* inherently strives for the dynamic balance; any abnormal phenomena do not endure indefinitely. Instead, these anomalies will be rectified through systemic chain reactions that gradually promote the restoration of stability and equilibrium.

From a limit perspective, the dynamic equilibrium within the system of *wuxing* is characterized as a relative trend rather than an absolute condition. This system undergoes continuous fluctuations, which manifest as a cycle of relative balance, local imbalance, and restored balance. This cyclical process enables the system to progress toward an overall state of equilibrium, although it never reaches a permanent steady state. A notable example of local imbalance within the human body is the occurrence of diseases. Therefore, the fundamental principle of disease prevention and treatment is to enhance and stimulate the human body's intrinsic self-regulatory capacity within the system of *wuxing*, which facilitates the restoration of disordered *qi* and chaotic numbers to a state of order and harmony.

6. Acupuncture Methods Based on Mathematical Principles

Based on the aforementioned mathematical principles, traditional Chinese medicine has developed many medical methods. Chinese medical thought initially emphasized the number six, while natural philosophy focuses on the number five [17] (p. 143). However, the theory of *wuyun liuqi* integrates these two mathematical constructs. It is grounded on the principles of *yinyang* and *wuxing* and incorporates *tiangan dizhi* 天干地支 (the heavenly stems and earthly branches) to analyze the dynamic patterns and interrelationships among climate, phenology, and pathological conditions.

By employing the mathematical principles of dichotomy and trichotomy, traditional Chinese medicine conceptualizes the *wuyun liuqi* as two interrelated, cyclical systems of *qi*, which elucidate the trajectory of *qi* as it circulates through the realms of heaven, earth, and the human body. It is characterized by complex man-made structures, as well as the correspondence between spatial and temporal. Although the calculation process requires the utilization of various diagrams, verses, and charts, its fundamental mathematical principle is simple, which can be expressed as follows: "Heaven employs [the number] six to generate terms. The earth employs [the number] five to cause restraint." 天以六為節, 地以五為制 [1] (Vol. 2, p. 184).

It refers to the *tianliu diwu* 天六地五 (heaven six and earth five) instead of the *tianwu diliu* 天五地六 (heaven five and earth six), which is rooted from the *dayan zhishu* in the *Yi Jing* and represents the derivation of the mathematical principles associated with the numbers of *liangyi and sansheng*. Conversely, the concept of *tianliu diwu* introduces mathematical principles of deformation that are grounded in binary and ternary divisions, which adds the dimensions of constant change to the mathematical framework.

The significance of *tiangan dizhi* in the theory of *wuyun liuqi* diverges markedly from their traditional pairing relationships, which have undergone a process of deformation. The primary cause of this deformation can be attributed to numerical discrepancies. The *wuyun* should correspond with the twelve *dizhi*, while the *Liuqi* should align with the ten *tiangan*. However, the misalignment introduces complexities in their pairing, since neither set is a multiple of the other. In order to solve this issue, traditional Chinese medicine splits the *huo* 火 (fire) in the *wuxing* into the *junhuo* 君火 (ruling fire) and *xianghuo* 相火 (ministerial fire), and makes special what is artificially stipulated, just as the medical expert in the Tang Dynasty Wang Bing (王冰 710–805) annotated: "*tianqi* 天氣 (the vital energy of heaven) does not correspond to the *junhuo*" 天氣不臨君火 [18] (p. 571). Additionally, the *qi* of *taiyin* 太陰 is assigned to the earthly branches of *chen* 辰, *xu* 戌, *chou* 丑, and *wei* 未, potentially due to the fact that it corresponds to the *tu* 土 (earth) in the *wuxing* and the central number of five, both of which possess particular significance. These modifications exemplify the mathematical principles of artificial constructs, such as transform the number five into six, or leave one position vacant.

Many acupuncture methods are intricately linked to *tiangan dizhi*, and they also employ the artificially constructed deformation mathematics due to numerical discrepancies. A typical example is the *linggui bafa* 靈龜八法 (the eight methods of an intelligent turtle), which is a method of selecting the acupuncture points based on time. Its theoretical basis is the numerology and patterns of the Eight Trigrams and Nine Palaces. Traditional Chinese medicine practitioners believe that this method can help them to observe and summarize the temporal patterns of the circulation of *qi* and blood in the human body, and then to determine the acupuncture point more accurately [19].

Traditional Chinese medicine believes that the *qi* and blood in the human body undergo spatial shifts over time. However, the meridians and acupuncture points through which *qi* and blood circulate and converge exhibit a relatively fixed spatial distribution. Consequently, one of the fundamental prerequisites for precise treatment involves the stimulation of various acupuncture points at different intervals, thereby effectively utilizing temporal factors to calculate spatial distribution. The *linggui bafa* primarily focuses on eight acupuncture points, whereas another acupuncture method of *ziwu liuzhu* 子午流注針法

(acupuncture based on the midnight–noon cycle of *qi*) encompasses as many as 66 acupuncture points. Although there is a significant difference in complexity and amount of calculation between these two methods, their foundational mathematical principles remain consistent, which is to integrate and harmonize *tiangan dizhi* with the meridians and acupuncture points through numerical alignment [19,20]. It facilitates the calculation of spatio-temporal variations in *qi* and the identification of suitable acupuncture points. Consequently, it enables the selection of points based on temporal considerations while concurrently adjusting *qi* through precise needling techniques.

Taking the *Linggui Bafa* as a case study, it focuses on eight acupuncture points where the *qijing bamai* 奇經八脈 (eight extraordinary meridians) intersect with the *shier zhengjing* 十二正經 (twelve primary meridians). Utilizing the relevant mathematical verses and diagrams, the mathematical formula for identifying the acupuncture points can be expressed as follows: (day stem-branch plus hour stem-branch) divided by 9 or 6 (divided by 9 if the day belongs to *yangri* 阳日. Conversely, it is divided by 6 if the day belongs to *yinri* 阴日). For instance, if a patient requires acupuncture during the hour of *renwu* 壬午时 and day of *yichou* 乙丑日, then the relevant verse should be firstly checked to determine the numbers to which they correspond, which are as follows: *ren* 壬 corresponds to 6, *wu* 午 corresponds to 9, *yi* 乙 corresponds to 9, *chou* 丑 corresponds to 10. And then the sum of them is 34. Since the *yichou* day belongs to *yinri*, which corresponds to the number 6, so 34 should be divided by 6 and the remainder is 4. By substituting the number 4 to the diagram of *jiugong*, the doctor can finally confirm that the accurate acupuncture point is *linqi xue* 臨泣穴 [2].

The essence of the *linggui bafa* is deeply rooted in the integration of the *qijing bamai*, *houtian bagua* 後天八卦 (the later heaven eight trigrams), and *mingtang jiugong* 明堂九宫 (the nine palaces of Mingtang). When the sequence and orientations of the *houtian bagua* are arranged within a nine-square grid, it can be found that the numbers and positions align precisely with the *mingtang jiugong*. The book of *Da Dai Li Ji* 大戴禮記 records a set of mysterious numbers related to the *mingtang* as "2, 9, 4, 7, 5, 3, 6, 1, 8" [21] (p. 150), which has been interpreted by many ancient Chinese ideologists. For example, Zhu Xi (朱熹 1130–1200), the preeminent Neo-Confucian master of the Song Southern Dynasty, described them as follows: "2 and 4 are the shoulders, 6 and 8 are the feet, 3 belong to the left and 7 belong to the right, 9 is above and 1 is below." 戴九履一, 左三右七, 二四为肩, 六八为足 [22] (p. 13).

The sum of the numbers in each vertical, horizontal, and diagonal line totals 15, which is a multiple of three, so it can be called the "Triple three magic square." Additionally, 15 is also a multiple of five, which serves as the central and significant number in the *mingtang jiugong*. The mathematical principles in the *Yi Jing* may be intricately connected to the central number five, which implies that the fundamental mathematical principle underlying the *mingtang jiugong* is also probably centered on five.

The distinctive treatment of the central number five within the *linggui bafa* also reflects its significance and singularity. In order to reconcile the numerical discrepancies among the *qijing bamai*, the Eight Trigrams, and the Nine Palaces, traditional Chinese medicine introduces the concept of *zhaohai kunerwu* 照海坤二五, which links the central number five and the special number two both to the acupuncture point of *zhaohai xue* 照海穴 [23] (p. 233). Subsequently, traditional Chinese medicine establishes connections between the central number five and the acupuncture points of *zhaohai xue* and *gongsun xue* 公孫穴, which correspond to the *kun* trigrams of *kun* 坤 and *qian* 乾respectively. If the remainder is 5 and the patient is male, the acupuncture point of *gongsun xue* is employed. Conversely, for a female patient, the acupuncture point of *zhaohai xue* is utilized.

7. Conclusions

The mathematical principles underlying the numbers represent a crucial connection that integrates the heaven, earth, human, *yinyang* and *wuxing* into a coherent system. Traditional Chinese medicine not only uses numbers to measure, analyze, and symbolize the human body's organs, data, and structures, but also views the mathematical principles as the crucial role for understanding the laws of life, developing medical theories, and executing medical practices.

The Chinese mathematical philosophy reflected in the body dimension are fundamentally anchored in the *zhishu*, which begin with the number one and end with the number nine. It is a nine-digit numerical system and spawns in two distinct development trajectories, one that advances towards simplified foundational mathematics, and another that evolves towards more complex artificial constructs. Both trajectories have significantly contributed to develop the theories and methods of traditional Chinese medicine, including the theories of *yinyang*, *wuxing*, and *wuyun liuqi*, as well as the acupuncture methods of *linggui bafa* and *ziwu liuzhu*.

Among the *zhishu*, the numbers one, two, and three embody the most special and important mathematical principles. Number one contains the mathematical principle of *hundun benglie*, and serves as the foundation of the mathematical structure of the human body. Number two contains the mathematical principles of *liangyi* and the bisection method of *yinyang* which have the characters of infinite divisions and symmetrical coupling. Number three contains two mathematical principles of *he'er weiyi* and *yifen weisan*, which is exemplified by the cosmic perspective of the correspondence between heaven, earth, and human. In addition, as ancient Chinese philosophers refined the theory of *wuxing*, traditional Chinese medicine also integrated them with *yinyang* and widely applied the mathematical principles underlying them in the medical theories and methods, thereby reinforcing the significance and uniqueness of the number five and its mathematical principles.

In the current context of artificial intelligence, it is essential to pay more attention to the mathematical philosophy that may be related to the foundational thought of traditional Chinese medicine. This approach may offer a broader perspective to comprehensively understand the traditional Chinese medicine in a more objective manner.

Author Contributions: Conceptualization, H.X. and R.Y.; methodology, H.X.; writing—original draft preparation, H.X.; writing—review and editing, H.X. and R.Y.; funding acquisition, H.X. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by China Postdoctoral Science Foundation, grant number 2024M753004, and National Social Science Fund of China, grant number 23FZXB028. And The APC was funded by 23FZXB028.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Conflicts of Interest: The authors declare no conflicts of interest.

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